AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_6C1BC7AF8D
Taxa	Apatosaurus
Clade	Sauropod
PBDB Records	67
Mass (regression)	7964.9 kg (Campione & Evans 2012)
Mass (volumetric)	7103.8 kg (Silhouette integral, ρ=1060.0 kg/m³)
Body Volume	8.457 m³
Validity Score	1.0000 / 1.0000
TCI	0.7003 (partial)
Axiom I Entropy	0.9585
Axiom II λ■	0.1805 (symmetry 100%)
Axiom III Uniformity	0.6637
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	66.7% Measured / 33.3% Inferred
Elements	27
CoG (m)	[0.891, 1.806, 0.0]
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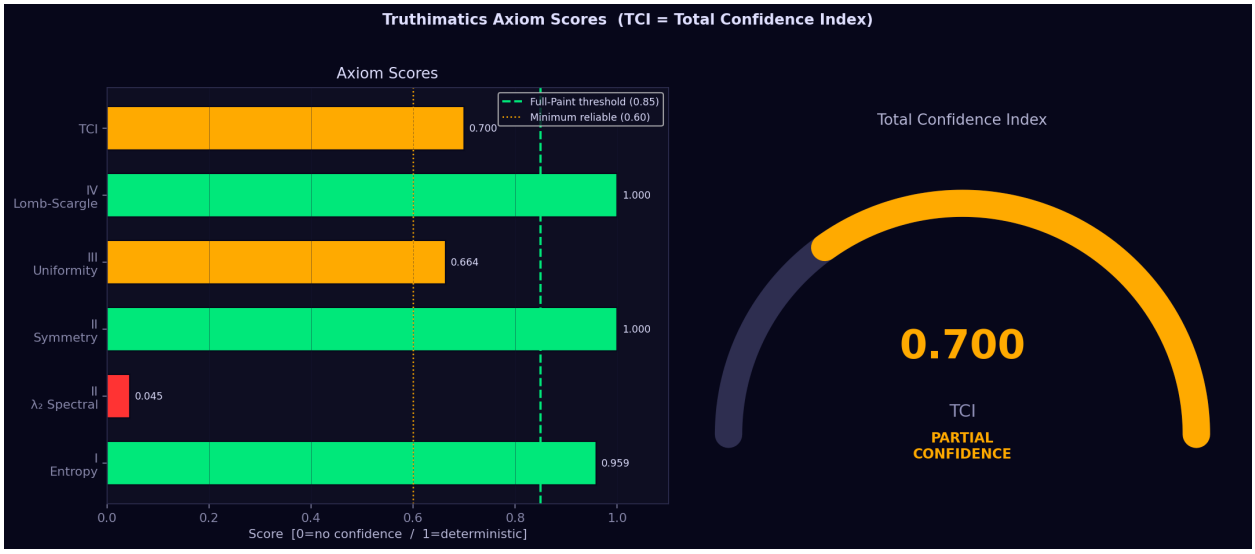
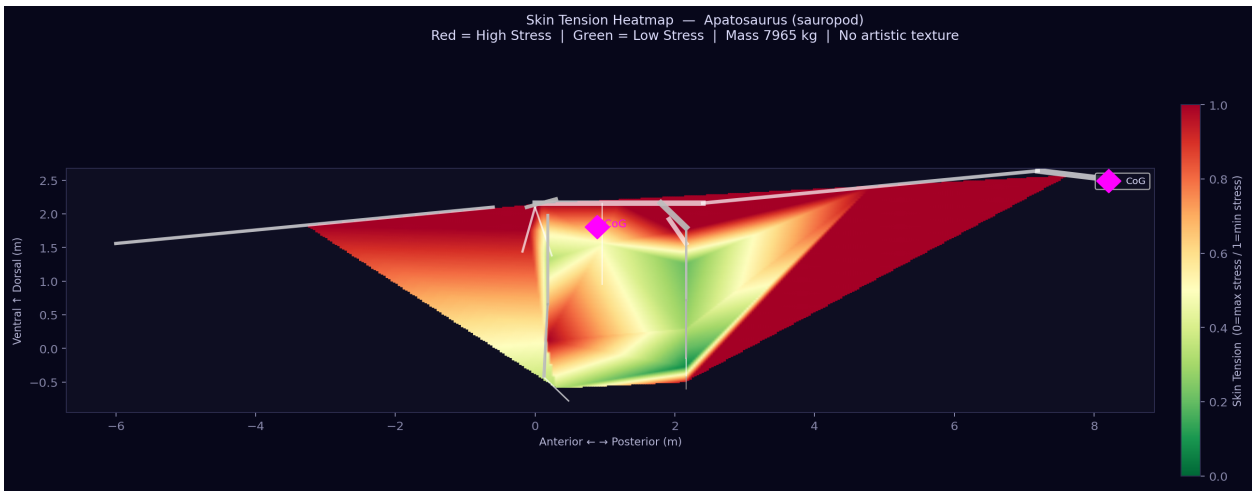
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

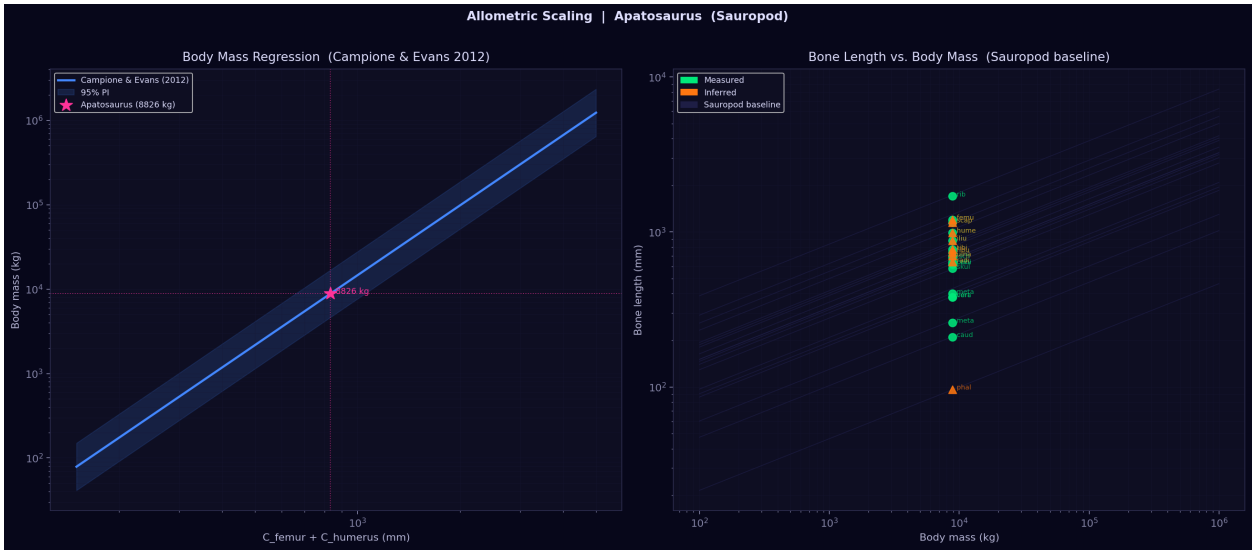
Loading: quadruped (F=Mg/4). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
caudal_vertebra	0.621	1.049	1.670	15.000	OK
cervical_vertebra	0.524	2.403	2.927	15.000	OK
coracoid	0.312	0.675	0.986	15.000	OK
femur	2.438	27.598	30.036	5.660	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	3.560	40.168	43.728	3.888	WARNING
ilium	1.420	7.196	8.616	15.000	OK
ischium	1.357	8.777	10.134	15.000	OK
metacarpal	14.103	67.526	81.629	2.083	WARNING
metatarsal	9.114	21.273	30.386	5.595	OK
phalanx	0.000	0.000	0.000	15.000	OK
pubis	2.317	17.861	20.178	8.425	OK
radius	2.804	30.674	33.478	5.078	OK
rib	0.371	21.555	21.926	7.753	OK
scapula	0.165	0.785	0.950	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	3.876	17.038	20.915	8.128	OK
ulna	2.483	27.971	30.454	5.582	OK
vertebra	0.654	1.214	1.868	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	2.37 m	Femur + tibia + metatarsal
Estimated Speed	11.9 km/h (3.3 m/s)	Gait: Walk | Fr=0.469
Stride Length	4.334 m	Alexander (1976) / Thulborn (1990)
Bite Force	2366.3 N (241.2 kgf)	Sauropod regression (Whitlock 2011)
Skull Length	580.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.3796	Circumference / length (dimensionless)
Limb Strength Index	3.3e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	16.74 m	Allometric from femur length
Est. Neck Length	7.03 m	Allometric from femur length
Est. Tail Length	7.03 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
caudal_vertebra	210.0	170.0	2430.96	3034.6	Measured	0.88	15.00
cervical_vertebra	620.0	185.0	8499.54	10610.0	Measured	0.88	15.00
coracoid	380.0	240.0	8767.31	10944.2	Measured	0.88	15.00
femur	1200.0	145.0	10105.95	12615.3	Measured	0.88	5.66
femur_right	1200.0	145.0	10105.95	12615.3	Symmetry	0.84	5.66
fibula	740.0	52.0	565.76	593.4	Measured	0.88	15.00
fibula_right	740.0	52.0	565.76	593.4	Symmetry	0.84	15.00
humerus	990.0	120.0	5710.28	7128.1	Measured	0.88	3.89
humerus_right	990.0	120.0	5710.28	7128.1	Symmetry	0.84	3.89
ilium	880.0	190.0	12724.77	15884.3	Measured	0.88	15.00
ilium_right	880.0	190.0	12724.77	15884.3	Symmetry	0.84	15.00
ischium	680.0	115.0	3602.17	4496.6	Measured	0.88	15.00
metacarpal	260.0	60.0	139.68	114.9	Measured	0.88	2.08
metatarsal	400.0	75.0	901.24	1125.0	Measured	0.88	5.59
phalanx	96.4	52.5	32.01	24.8	Constraint	0.40	15.00
pubis	620.0	88.0	1923.17	2400.7	Measured	0.88	8.43
radius	640.0	80.0	1640.67	2048.0	Measured	0.88	5.08
radius_right	640.0	80.0	1640.67	2048.0	Symmetry	0.84	5.08
rib	1700.0	48.0	1568.89	1958.4	Measured	0.88	7.75
scapula	1150.0	330.0	50163.26	62618.8	Measured	0.88	15.00
scapula_right	1150.0	330.0	50163.26	62618.8	Symmetry	0.84	15.00
skull	580.0	330.0	25299.73	31581.7	Measured	0.88	15.00
tibia	770.0	115.0	4078.93	5091.7	Measured	0.88	8.13
tibia_right	770.0	115.0	4078.93	5091.7	Symmetry	0.84	8.13
ulna	700.0	85.0	2025.80	2528.8	Measured	0.88	5.58
ulna_right	700.0	85.0	2025.80	2528.8	Symmetry	0.84	5.58
vertebra	380.0	280.0	11933.28	14896.3	Measured	0.88	15.00